SINGULAR FIBRES OF AN ORTHOCENTER SLIDING WINDOW OVER ANISOTROPIC FINITE PLANES

ANONYMOUS

Abstract. The orthocentric-quartet recurrence is classical, and metrical triangle geometry over finite fields is established; both receive zero contribution credit here. We totalize the sliding window $(A,B,C) \mapsto (B,C,H(A,B,C))$ over $\mathbb{F}_p^2$, for primes $p \equiv 3 \pmod 4$, by sending a degenerate next window to a sink. The residual singular dynamics has a complete orientation-sensitive atlas. Nonright triangles form exact four-cycles. A triangle right-angled at its first listed vertex has depth two, whereas right angles at the second or third vertex have depth one. We count all three strata, identify the stable image, and give the fibre of every target: zero over the first two oriented right-angle classes, one over the third and over every nonright triangle, and $1 + 2p^2(p^2-1)(p-1)$ over the sink. At $p = 3$ the periodic triangle core is empty but the height remains two. Exact enumeration at $p = 3, 7$ checks 1,317,843 assertions, including every state, target, and reverse window.

1. Owned geometry, totalized map, and theorem

Kocik and Solecki explicitly describe the four vertices of an orthocentric system, each point being the orthocenter of the triangle formed by the other three [2]. Wildberger develops affine metrical triangle geometry and orthocenters over finite fields [3]. Guy also treats a triangle as an orthocentric quadrangle, but his iterative "trisequence" is generated by Steiner-line reflections and circumcircle intersections, not by the sliding window below [1]. Accordingly, the orthocentric identity, its four-window recurrence, and the finite-field setting are background only.

Let $p \equiv 3 \pmod 4$ be prime. On $V = \mathbb{F}_p^2$ use

$$(x,y) \cdot (u,v) = xu + yv.$$

An ordered triple (A,B,C) is a triangle when its points are noncollinear. Its orthocenter $H(A,B,C)$ is the unique solution of

$$(H-A)\cdot(B-C) = 0, \qquad (H-B)\cdot(A-C) = 0. \tag{1}$$

Let $\mathcal{X}_p$ contain every ordered triangle and one sink $\dagger$. Define

$$F(A,B,C) = \begin{cases} (B,C,H(A,B,C)), & B,C,H(A,B,C) \text{ are noncollinear,} \\ \dagger, & \text{otherwise,} \end{cases} \qquad F(\dagger) = \dagger. \tag{2}$$

A triangle is *right at coordinate i* when its two side vectors from the ith listed vertex are perpendicular. Put

$$T = p^2(p^2-1)(p^2-p), \quad R = p^2(p^2-1)(p-1), \quad Q = T - 3R = p^2(p^2-1)(p-1)(p-3). \tag{3}$$

For a state x, let $d(x)$ be its distance to the recurrent set.

Theorem 1 (singular and fibre atlas)**.** *The map F has the following complete description.*

(i) There are T ordered triangles and exactly R right triangles at each specified coordinate. The three right-angle classes are disjoint. The Q nonright triangles have exact period four; the sink is fixed.

(ii) A triangle right at coordinate one has depth two. A triangle right at coordinate two or three has depth one. Hence the sharp height is two, and

$$\#\{x : d(x) \le 0\} = 1 + Q, \quad \#\{x : d(x) \le 1\} = 1 + T - R, \quad \#\{x : d(x) \le 2\} = 1 + T. \tag{4}$$

The dynamical zeta function is

$$\zeta_F(z) = (1-z)^{-1}(1-z^4)^{-Q/4}. \tag{5}$$

(iii) The one-step target fibres are

$$\#F^{-1}(y) = \begin{cases} 1+2R, & y = \dagger, \\ 0, & y \text{ is right at coordinate one or two}, \\ 1, & y \text{ is right at coordinate three or is nonright.} \end{cases} \tag{6}$$

Consequently

$$\operatorname{im} F = \{\dagger\} \cup \{\text{nonright triangles}\} \cup \{\text{triangles right at coordinate three}\}, \tag{7}$$

$$\#\operatorname{im} F = 1 + T - 2R, \qquad F^2(\mathcal{X}_p) = \{\dagger\} \cup \{\text{nonright triangles}\}, \tag{8}$$

$$\#F^2(\mathcal{X}_p) = 1 + Q, \qquad F^t(\mathcal{X}_p) = F^2(\mathcal{X}_p) \quad (t \geq 2). \tag{9}$$

(iv) At $p = 3$, $T = 432$, $R = 144$, and $Q = 0$. Thus the sink is the only recurrent state, while the depth-two shell still contains 144 *triangles. The one-step image has size* 145*, the stable image has size one, and the sink fibre has size* 289*.*

Table 1. Primary-source subtraction. The middle column receives zero contribution credit; the last column is the narrow residual.

Source	Owned input	Residual retained here
Kocik–Solecki [2]	orthocentric quartet and mutual-orthocenter identity	none of the four-window periodic mechanism
Wildberger [3]	affine metrical triangle geometry and orthocenters over finite fields	anisotropic singular totalization only
Guy [1]	orthocentric-quadrangle viewpoint and a different iterative construction	no claim against that Steiner/reflection trisequence
Elementary geometry	perpendicular counts, generic zeta conversion	oriented depth, every-target fibre, and stable-image conjunction

2. Anisotropy and the right-angle strata

Because $p \equiv 3 \pmod 4$, -1 is not a square. Therefore

$$x^2 + y^2 = 0 \quad \Longrightarrow \quad (x, y) = (0, 0). \tag{10}$$

Indeed, if $y \neq 0$ then $(x/y)^2 = -1$, and if $y = 0$ then $x = 0$. Thus the dot product is anisotropic. A nonzero vector (x, y) has perpendicular line spanned by $(-y, x)$, and no nonzero perpendicular vector is parallel to it.

There are p^2 choices for A, then $p^2 - 1$ choices for $B - A$, and $p^2 - p$ choices for C off the affine line AB. This proves the formula for T. To count triangles right at A, choose A, a nonzero vector $u = B - A$, and one of the $p - 1$ nonzero vectors $v = C - A$ on $u^\perp$. Anisotropy makes u, v independent, giving R.

The three right classes cannot overlap. For example, if $u = B - A$ and $v = C - A$ made the triangle right at both A and B, then

$$u \cdot v = 0, \qquad (-u) \cdot (v - u) = 0$$

would imply $u \cdot u = 0$, contradicting (10). The other pairs follow by relabeling. Subtracting the three disjoint classes proves the formula for Q in (3). The altitude equations (1) have a unique solution because the two opposite-side vectors are independent and the dot pairing is nondegenerate.

3. Four windows and oriented singular depths

We include an algebraic verification of the owned orthocentric identity only to close the field-specific proof. From (1), subtracting the two zero scalar products gives $(H - C) \cdot (A - B) = 0$. Thus H lies on all three altitudes. Conversely, the same three perpendicularities show that A lies on the altitudes of (B, C, H). Hence

$$H(B, C, H(A, B, C)) = A \tag{11}$$

whenever that next triangle is noncollinear; cyclic variants hold as well.

If the original triangle is nonright, H is distinct from A, B, C. No three of the four points are collinear: otherwise the altitude equations and anisotropy force the alleged third point to coincide with an endpoint. Equation (11) therefore produces the valid windows

$$(A, B, C),\ (B, C, H),\ (C, H, A),\ (H, A, B),\ (A, B, C). \tag{12}$$

No return at time one, two, or three is compatible with the three distinct ordered vertices. The period is exactly four. This classical core receives no contribution credit.

For a right triangle the orthocenter is its right-angle vertex. If it is right at A, the first successor is the cyclic rotation (B, C, A), which is right at its third coordinate; its next successor is degenerate and hence †. If the original triangle is right at B or C, the first next window already repeats that coordinate and goes to †. This proves the oriented depths, sharp height, stable recurrent set, and (4). There is one fixed cycle and $Q/4$ four-cycles, which gives (5) by the standard finite-map product formula.

4. The unique reverse window and all fibres

The fibre law is not obtained by dividing orbit counts. Let (A, B, C) be a target triangle. Any nonsink predecessor has the form (D, A, B) and must satisfy $H(D, A, B) = C$. Orthocentric symmetry then forces the unique candidate

$$(D, A, B) = (H(A, B, C), A, B). \tag{13}$$

If the target is right at A, its orthocenter equals A, so (13) repeats A and is invalid. If it is right at B, the candidate repeats B and is again invalid. For a target right at C, the candidate is the cyclic rotation (C, A, B) and maps to the target. For a nonright target, the four points in (12) are distinct with no collinear triple, so the candidate is valid and maps to the target. This proves the triangle cases of (6) directly.

The immediate predecessors of † are † itself and precisely the triangles right at coordinate two or three. These two disjoint classes have total size $2R$, proving the sink fibre. Reading the positive fibres gives (7); applying the oriented depth result once more gives (8). This completes Theorem 1.

5. Exact controls and claim ceiling

The paper-local verifier exhausts all 433 states at $p = 3$ and all 98,785 states at $p = 7$. It solves every orthocenter over the field, checks anisotropy and all altitude equations, classifies every oriented right angle, follows every orbit, reconstructs every reverse window, and compares all target fibres and images with Theorem 1. Its deterministic transcript contains 1,317,843 exact assertions. These computations are finite counterexample pressure; Sections 2–4 prove the all-prime result.

The theorem requires prime $p \equiv 3 \pmod 4$ and this dot product. It does not cover isotropic primes, arbitrary quadratic forms, higher dimensions, or perturbed centers. A bounded search did not locate the singular sliding- window fibre conjunction, but that non-hit is not a novelty, priority, ownership-completeness, or release certificate. The finite-field geometry, orthocenter, and orthocentric four-cycle remain zero-credit background.

Declarations

Data availability. No external data were used. The verifier and transcript contain the deterministic exact controls.

Ethics. This mathematical study involved no human participants, animals, personal data, or field intervention.

Author contributions. The anonymous author performed the derivations, exact checks, source-boundary audit, and manuscript preparation.

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External status. This artifact remains `HOLD_EXTERNAL`; it is not cleared for posting, submission, circulation, or author contact.

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